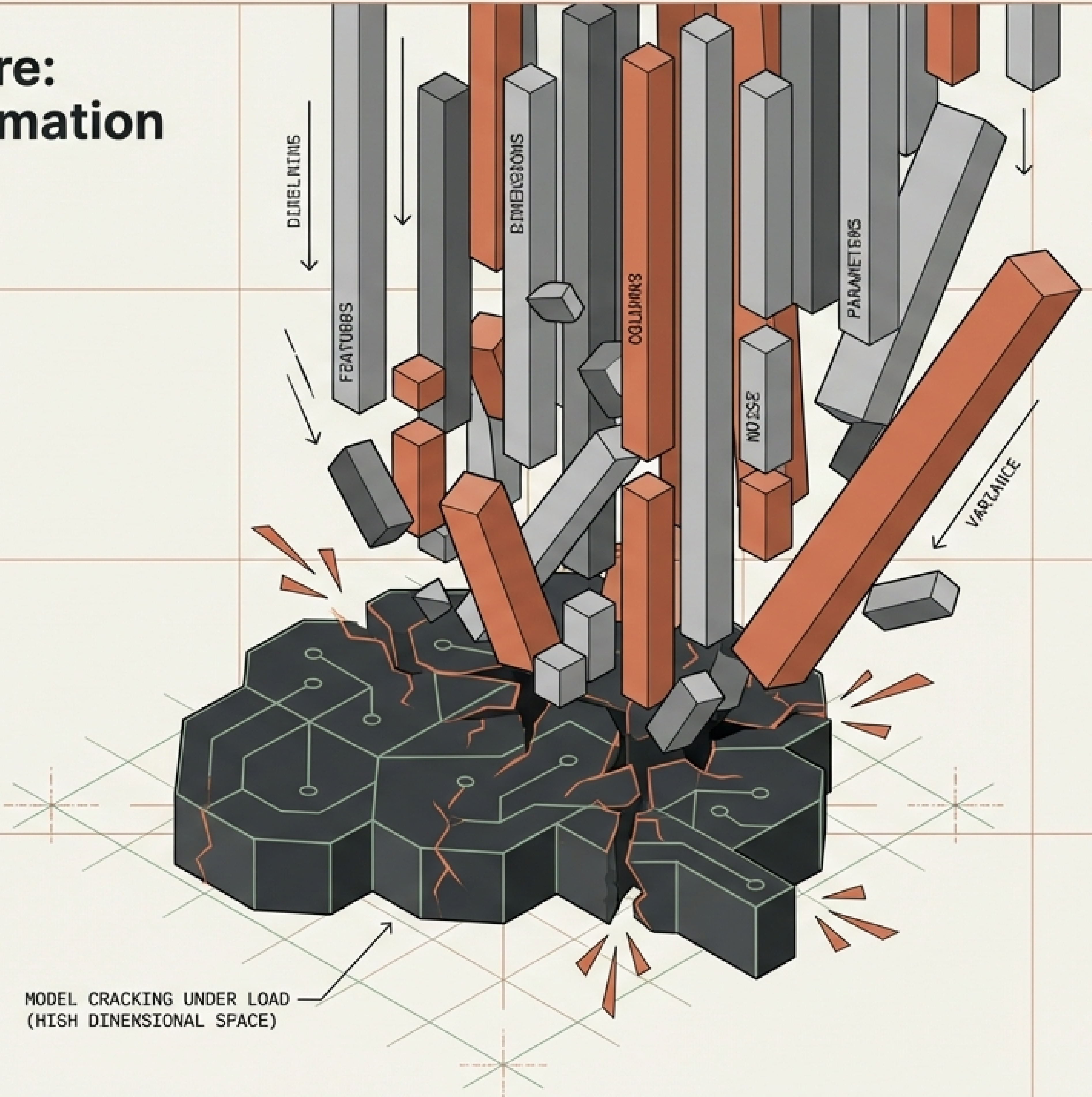
The Paradox of More: When Adding Information Breaks the Model

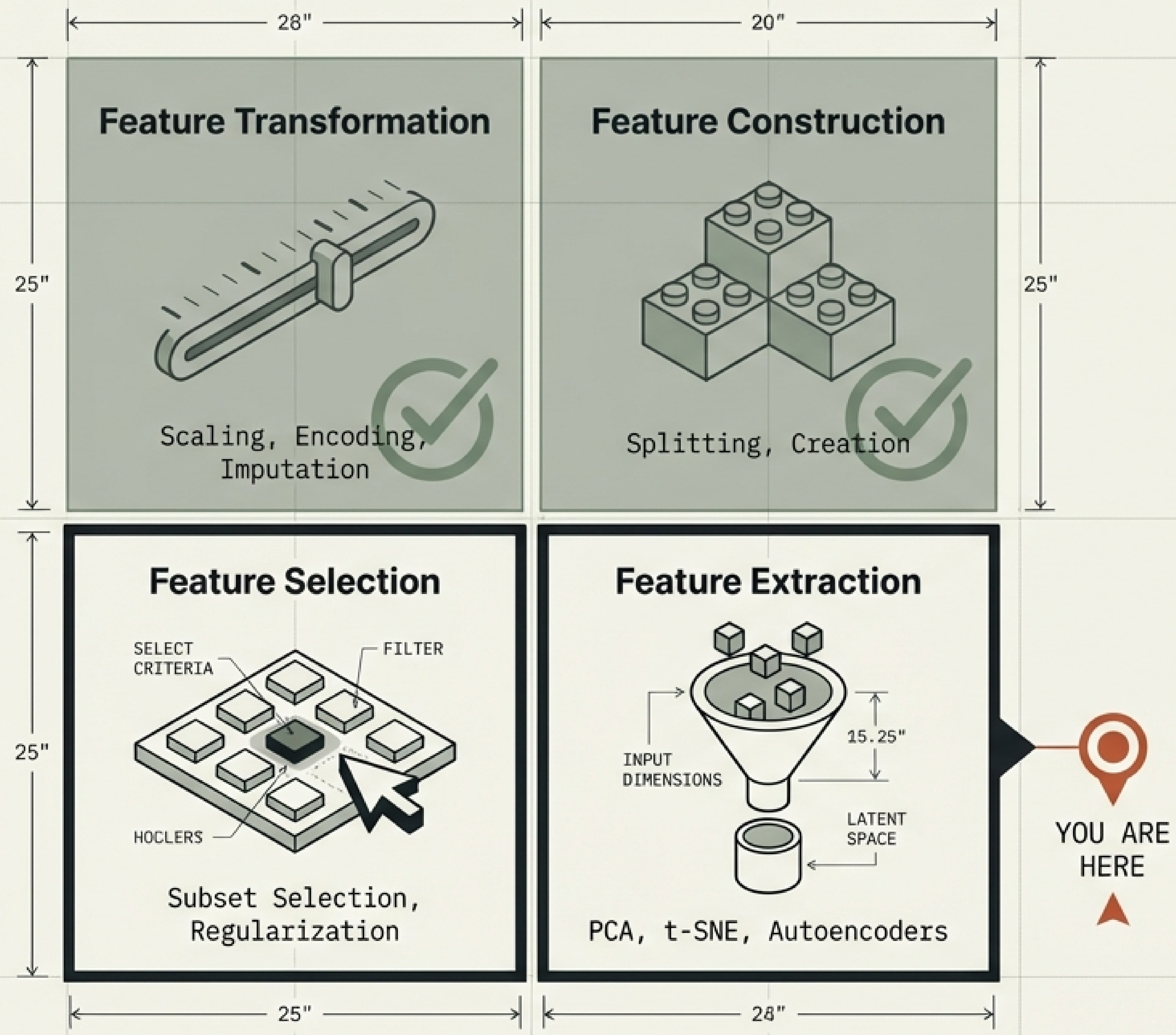
INTRODUCTION TO THE CURSE OF DIMENSIONALITY

In Machine Learning, we often assume that feeding more features (columns) to a model improves its accuracy. However, there is a tipping point where additional data creates noise rather than signal.



“THE CURSE OF DIMENSIONALITY”

TECHNICAL NOTE: AS DIMENSIONS INCREASE, THE VOLUME OF SPACE EAPAKBS EXPONENTIALLY, MAKING DATA SPARSE.
- DATA SCIENTIST REF. 1A



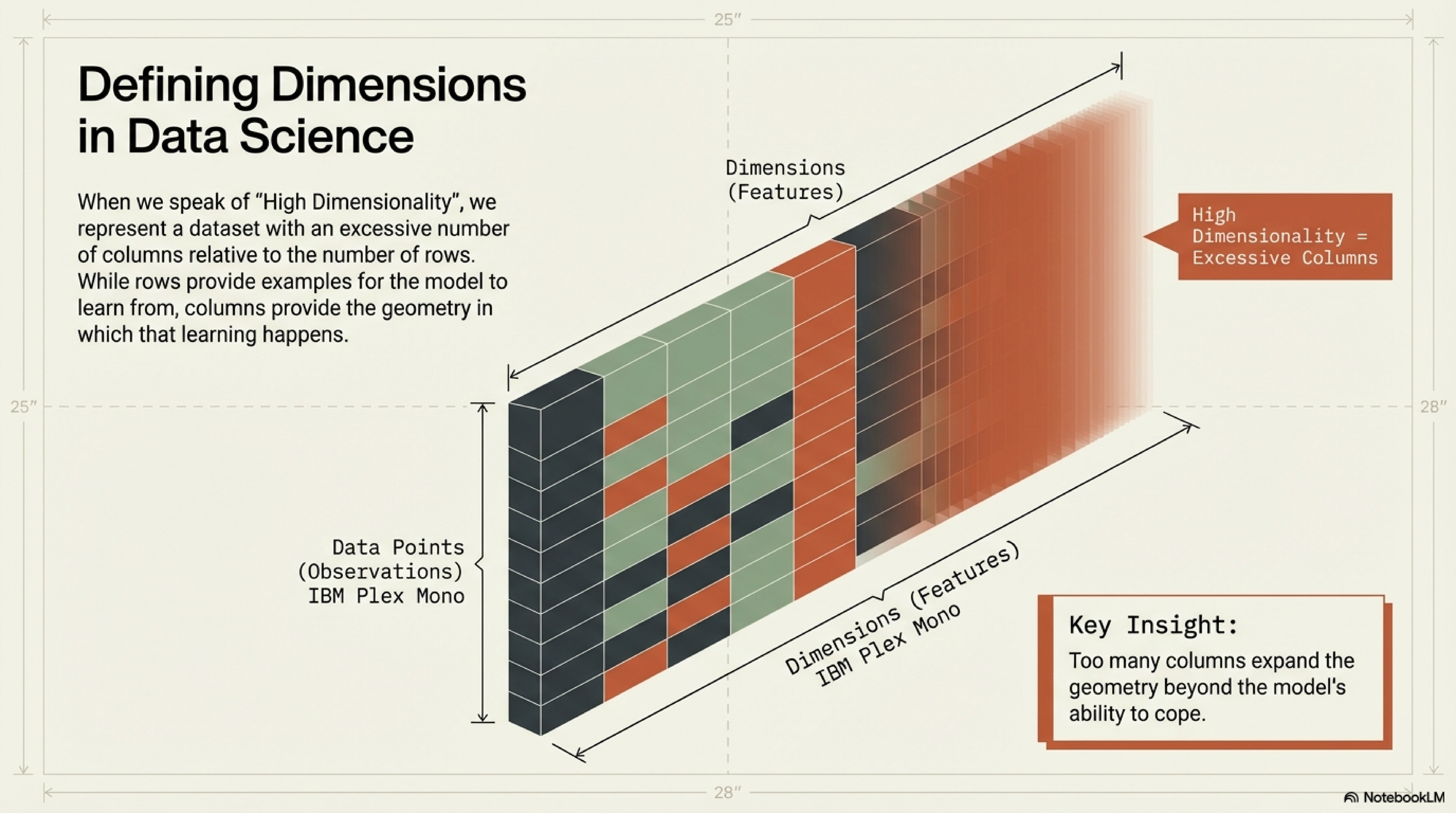
CURRICULUM ROADMAP IBM Plex Mono

Mapping the Feature Engineering Landscape

We have mastered cleaning and building features. Now, we move to the sophisticated task of managing quantity. Both Selection and Extraction are the direct answers to the Curse of Dimensionality.

Defining Dimensions in Data Science

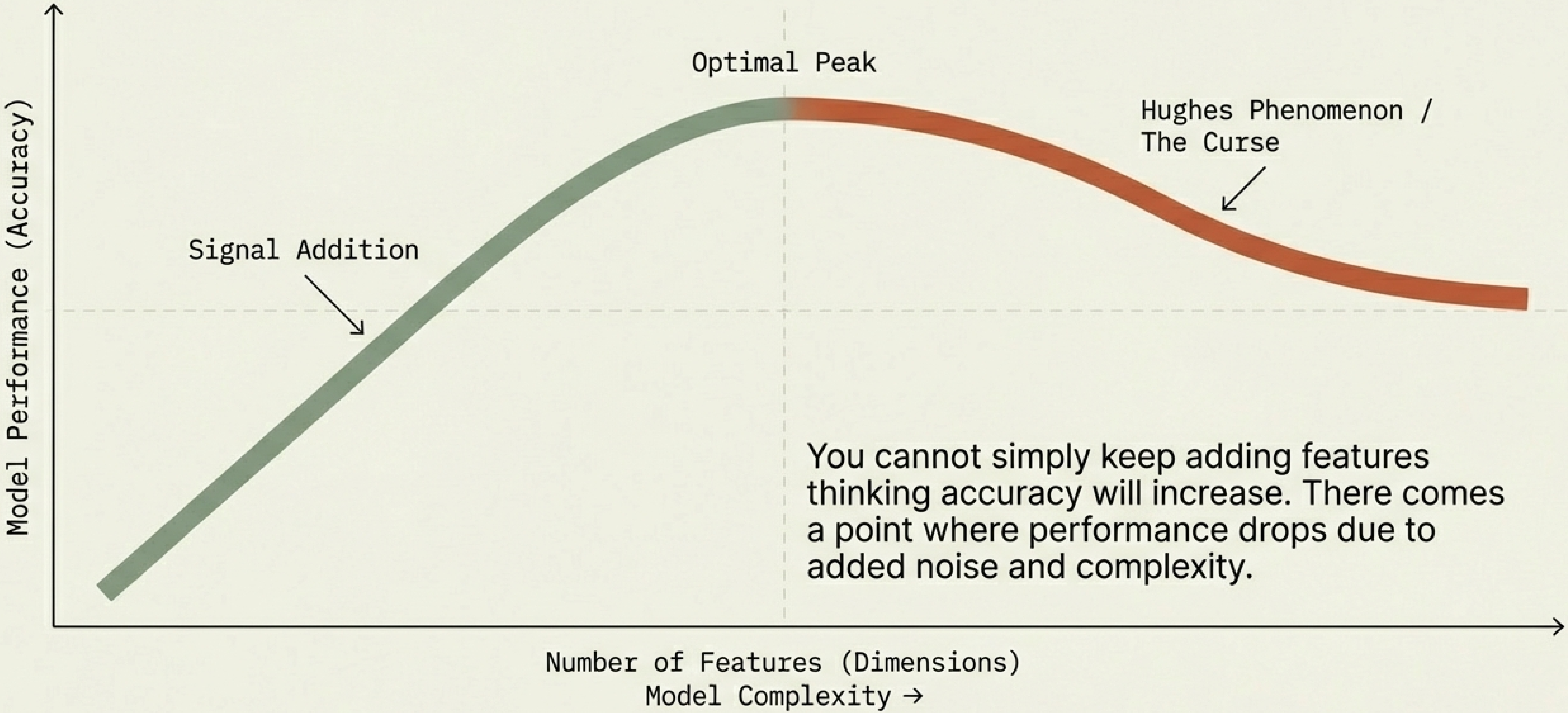
When we speak of "High Dimensionality", we represent a dataset with an excessive number of columns relative to the number of rows. While rows provide examples for the model to learn from, columns provide the geometry in which that learning happens.



Key Insight:

Too many columns expand the geometry beyond the model's ability to cope.

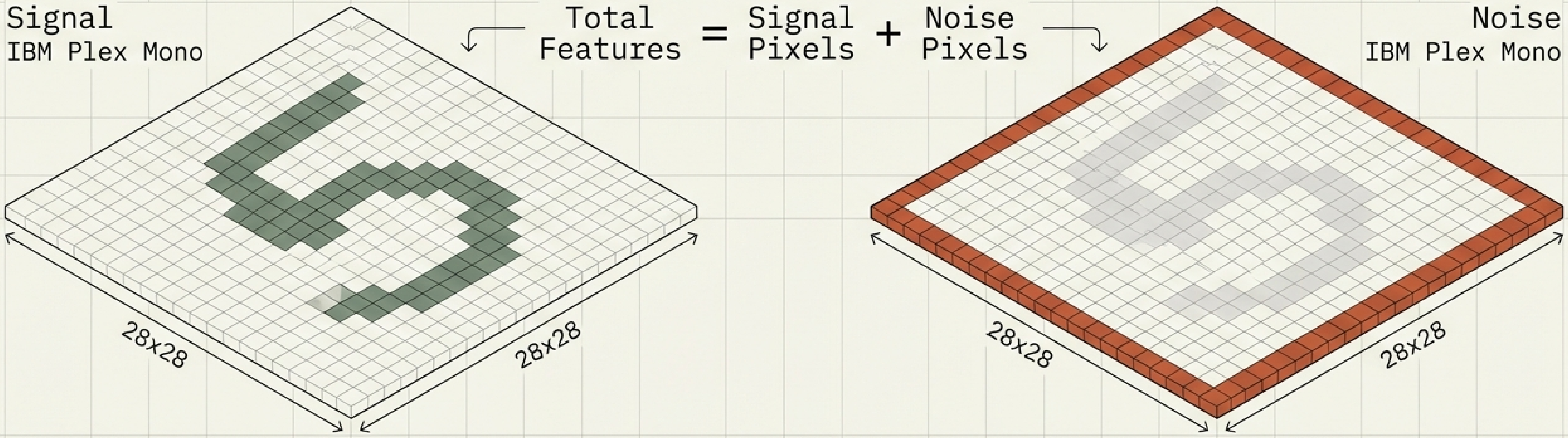
The Threshold of Diminishing Returns



You cannot simply keep adding features thinking accuracy will increase. There comes a point where performance drops due to added noise and complexity.

Case Study: Signal vs. Noise in Image Data

In an image, every pixel is a feature column. The central pixels contain the pattern (Signal). The border pixels are blank and do not change between digits (Noise). Feeding the border pixels creates unnecessary computational load and confusion.



The Analogy of the Lost Wallet: 1D and 2D Search

1D: The Road
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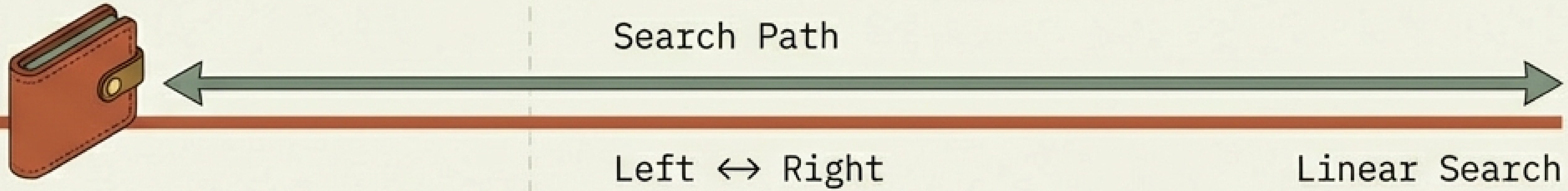

1 Dimension (Line): You dropped a wallet on a road.
You only search forward or backward.
Difficulty: Low.

2D: The Field
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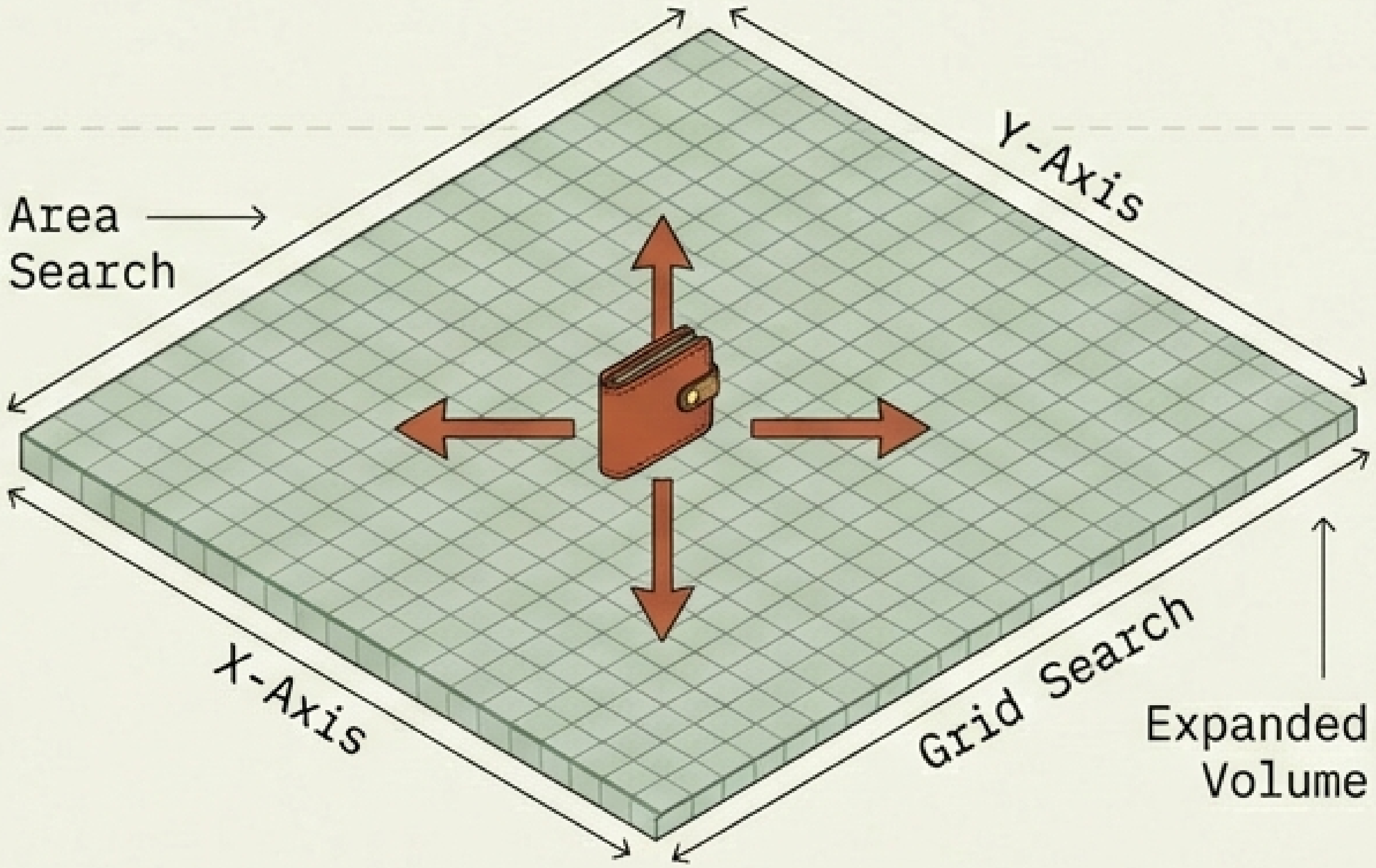
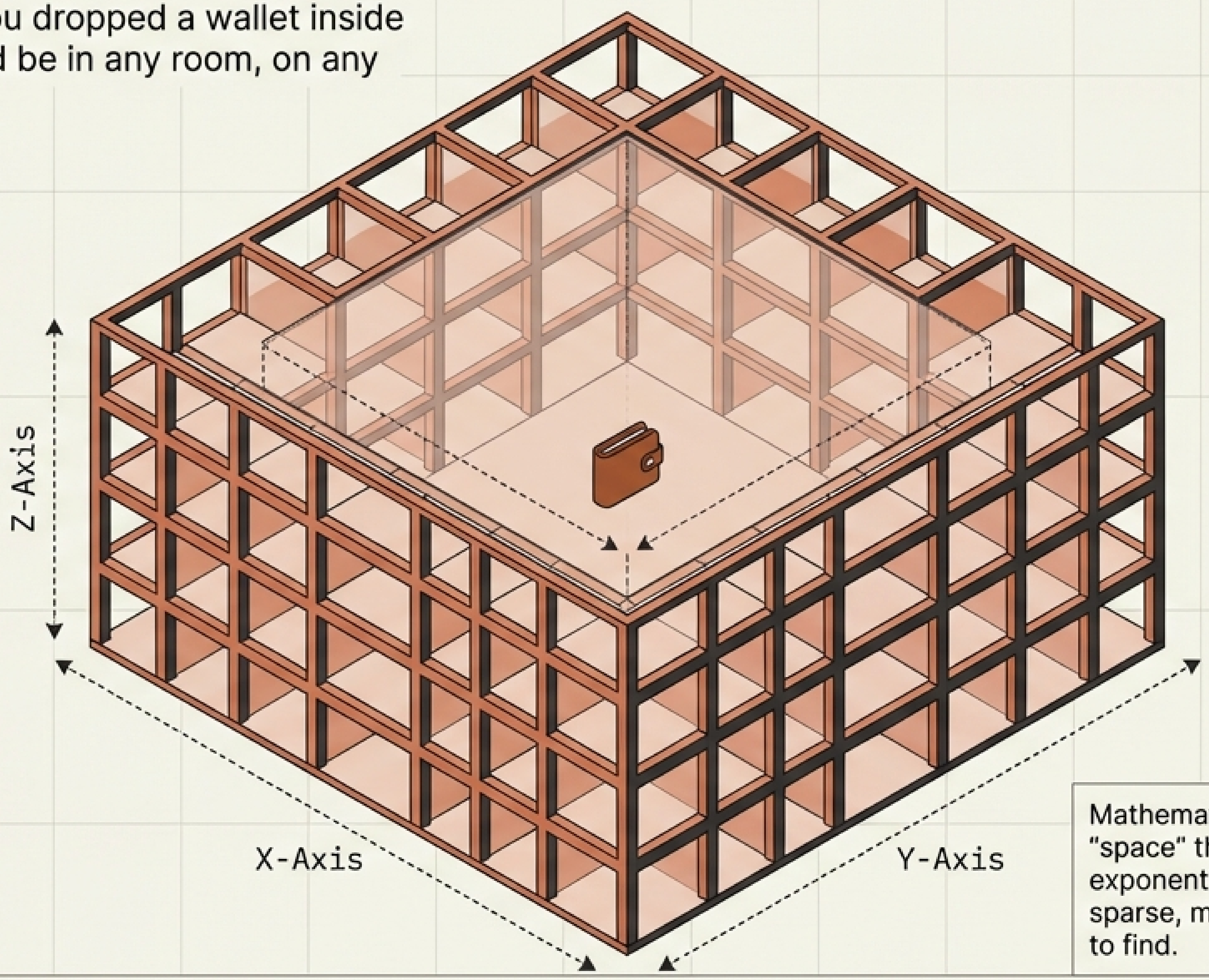


2 Dimensions (Plane): You dropped a wallet on a massive field. You must search a grid (X and Y axes). The search area has grown significantly. Difficulty: Moderate.

The Explosion of Space: Entering the 3rd Dimension

3 Dimensions (Volume): You dropped a wallet inside a massive building. It could be in any room, on any floor, or floating in the air.



Mathematical Reality: As dimensions rise, the "space" the model must cover grows exponentially. Data points become incredibly sparse, making the "wallet" nearly impossible to find.

The Mathematics of Sparsity

Helvetica Now Display

The Rule:

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As dimensions increase arithmetically, the volume of the space increases exponentially.

Consequence:

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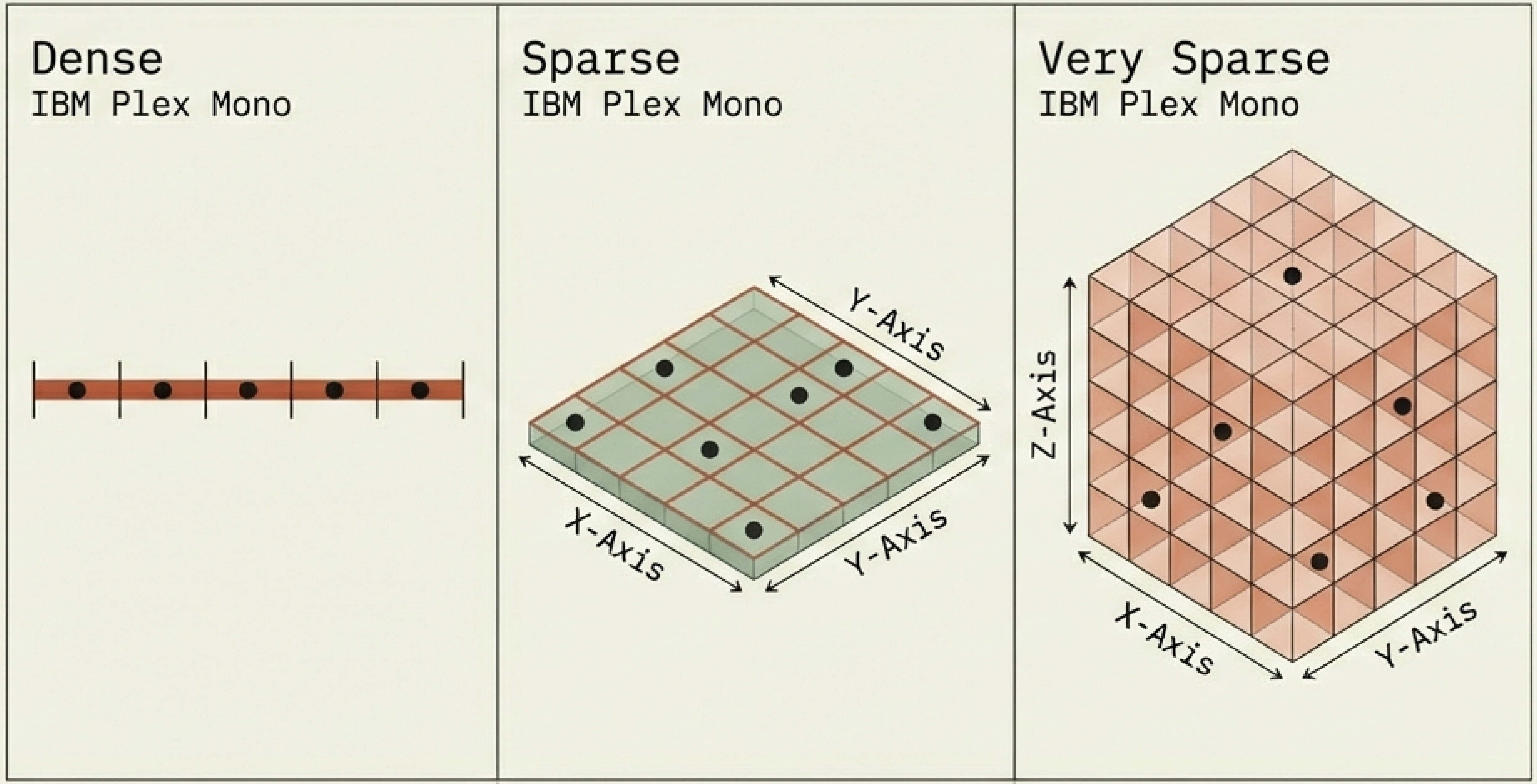
Unless you increase your training data exponentially to keep up, your dataset becomes empty. The model has “nothing to learn from” in the vast gaps between points.



Dense: 5 segments, 5 data points.

Sparse: 25 squares, 5 data points.

Very Sparse: 125 blocks, 5 data points.

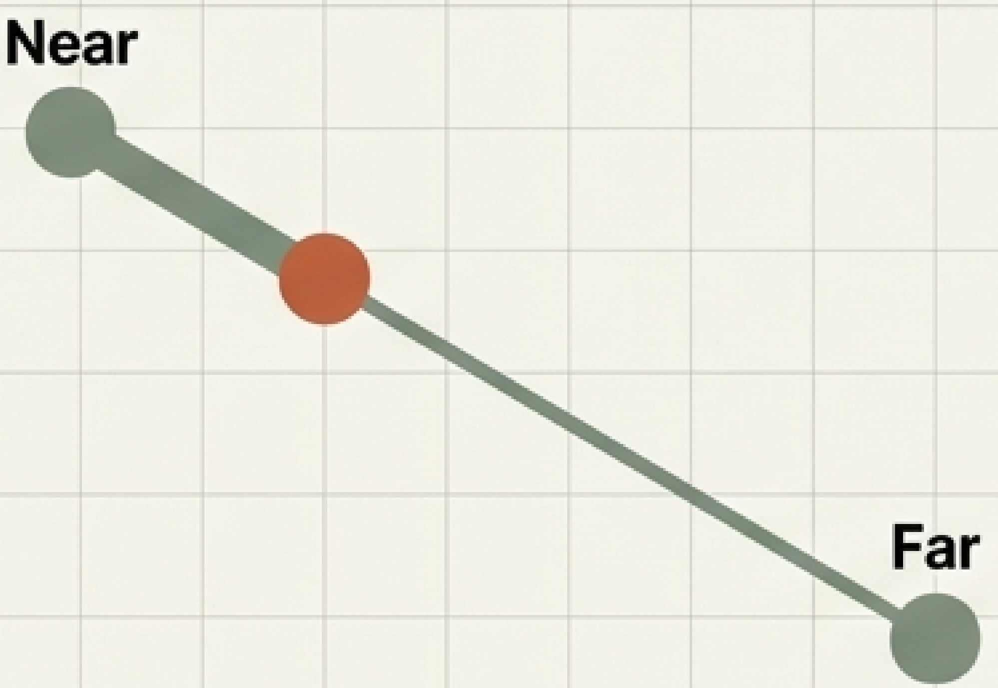
28"

When Distance Loses Meaning

Algorithms like K-Nearest Neighbours (KNN) rely on calculating the distance between points to classify them. In high-dimensional space, all points drift so far apart that the distinction between "near" and "far" vanishes.

The model becomes blind, unable to cluster similar items.

Low Dimension



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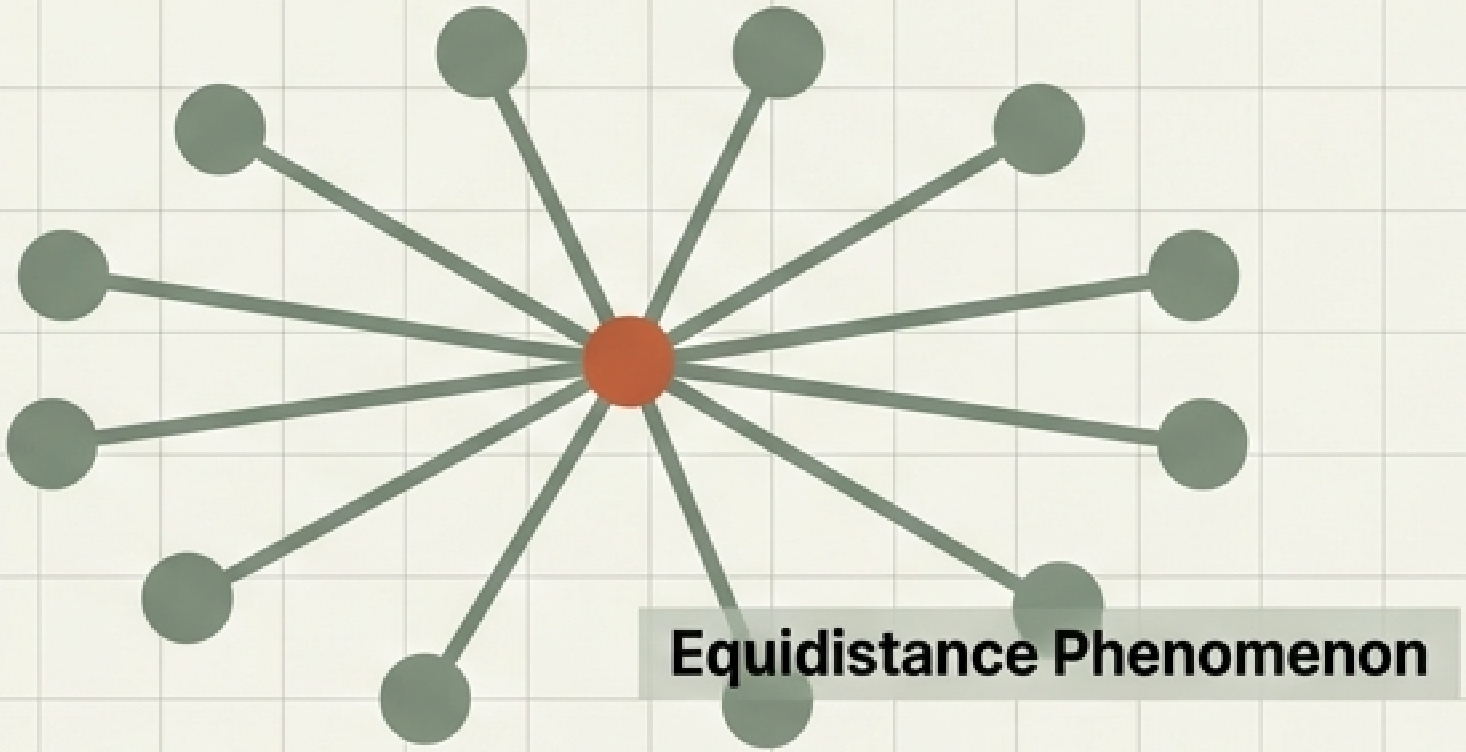
High Dimension



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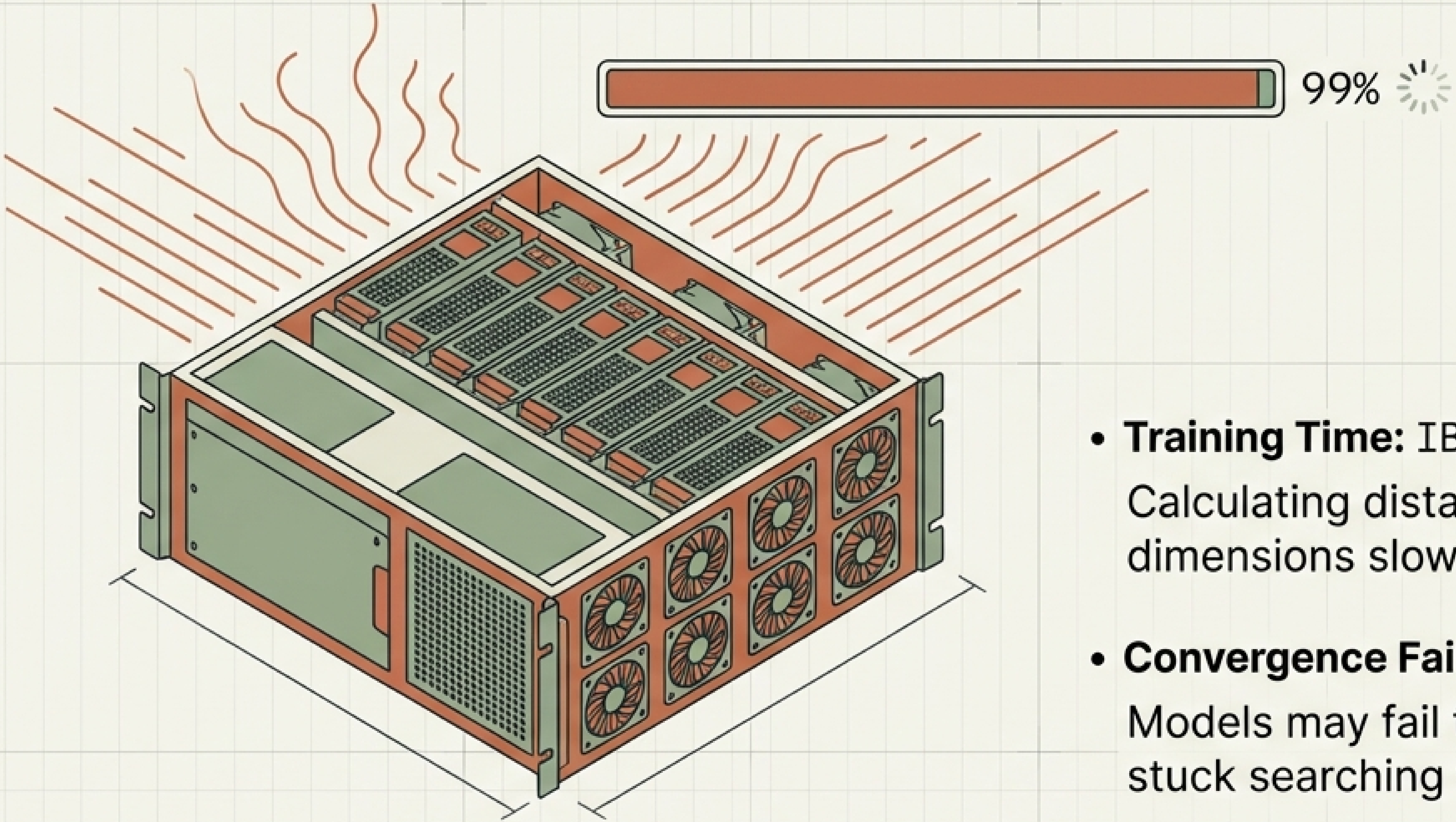
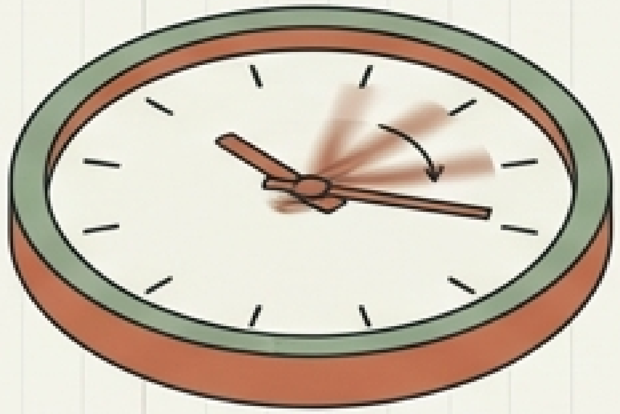
The Computational Penalty





- **Training Time:** IBM Plex Mono
  Calculating distances across thousands of dimensions slows training to a crawl.

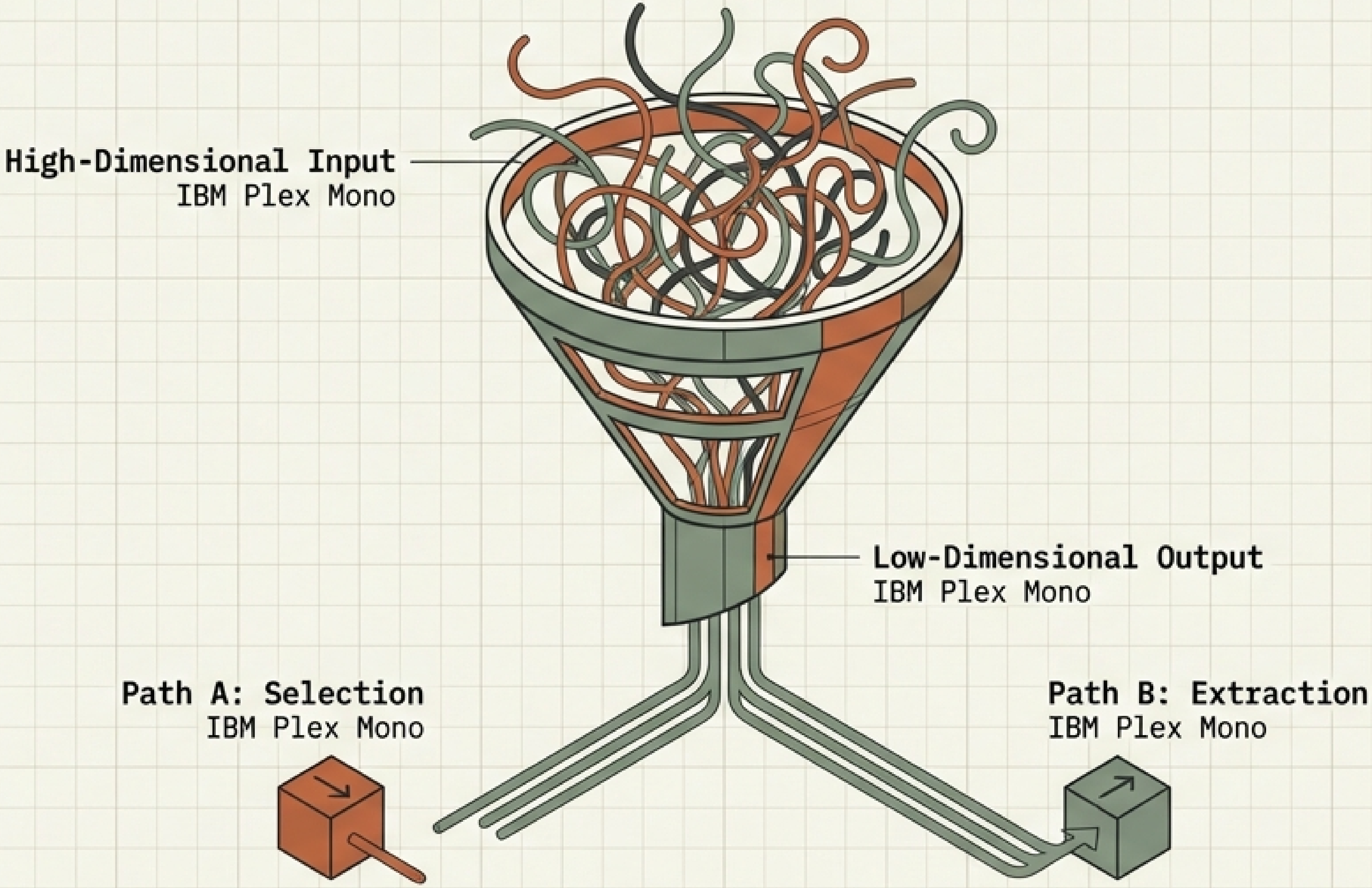
- **Convergence Failure:** IBM Plex Mono
  Models may fail to optimize entirely, getting stuck searching the vast empty space.

The Curse of Dimensionality is not just about accuracy; it is about efficiency and feasibility.

The Solution: Dimensionality Reduction

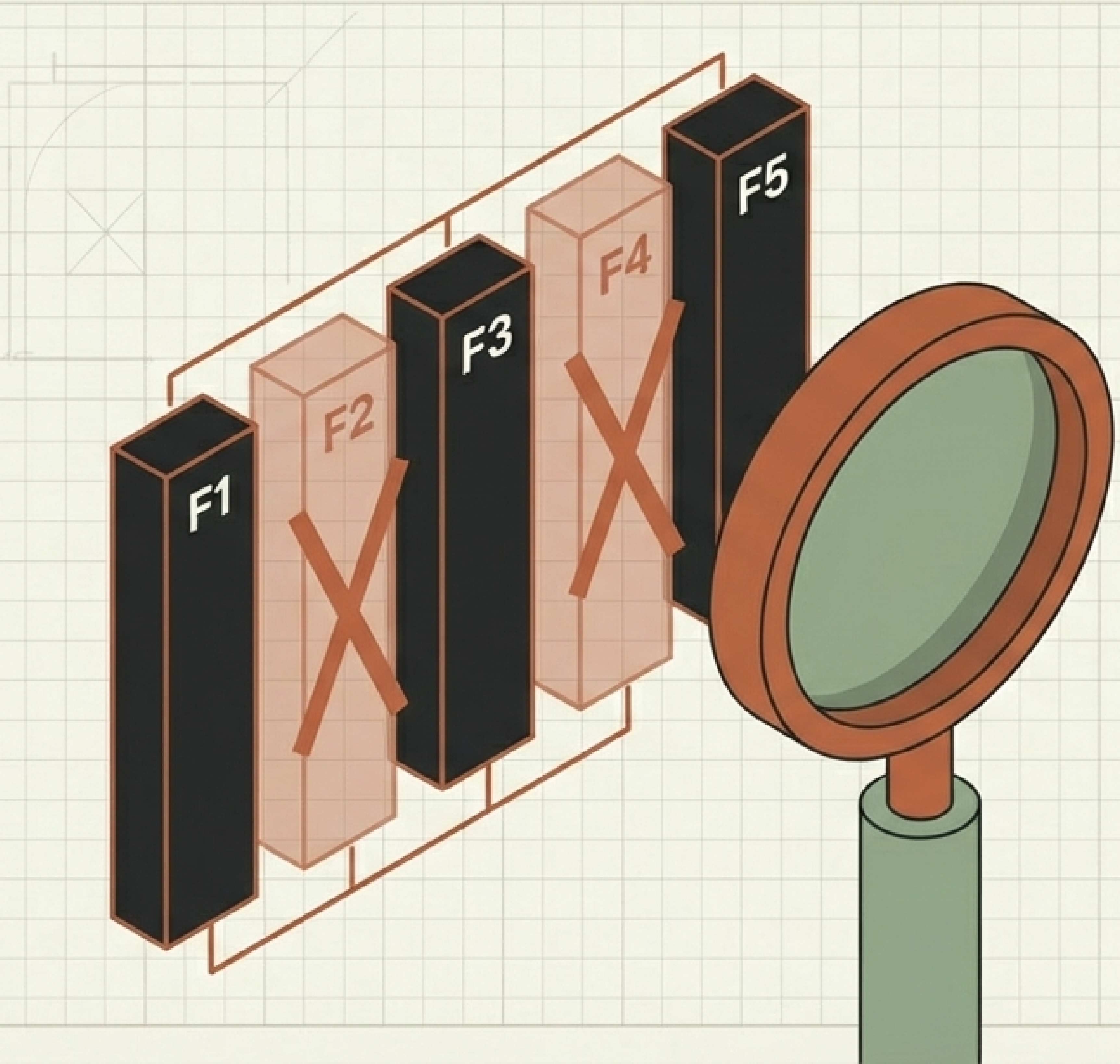
To save our model, we must reduce the number of features while preserving the essence (variance) of the data.

Two Strategic Paths



Approach 1: Feature Selection

Selecting the most important subset of the ORIGINAL features.



Techniques: `IBM Plex Mono`

1. Forward Selection
   (Add best one by one)

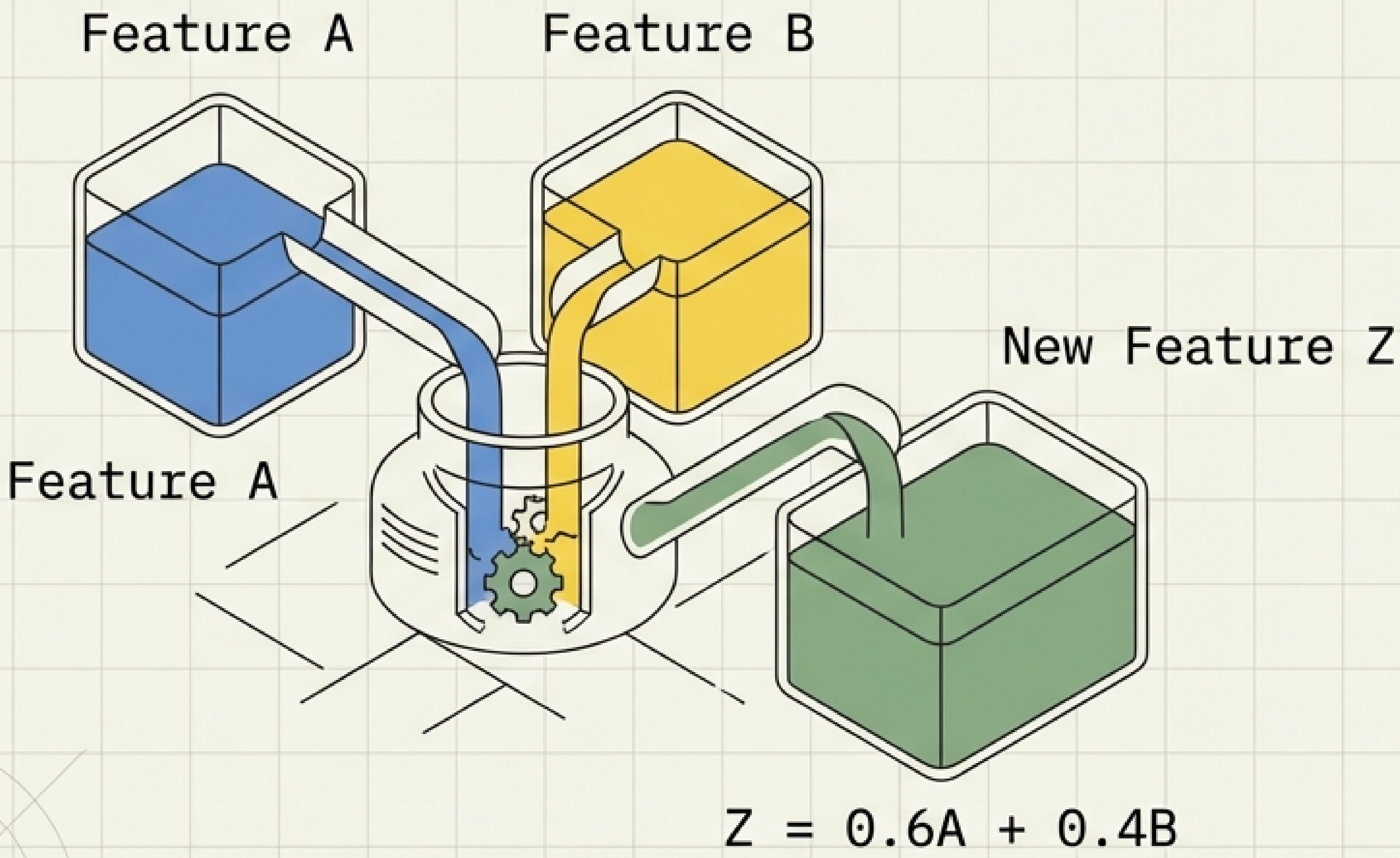
2. Backward Elimination
   (Remove worst one by one)

Benefit:

The features remain human-readable. You keep the original 'Age' or 'Salary' columns, just fewer of them.

Approach 2: Feature Extraction

Creating a **completely new, smaller set of features** that captures the information from the original set through mathematical projection.



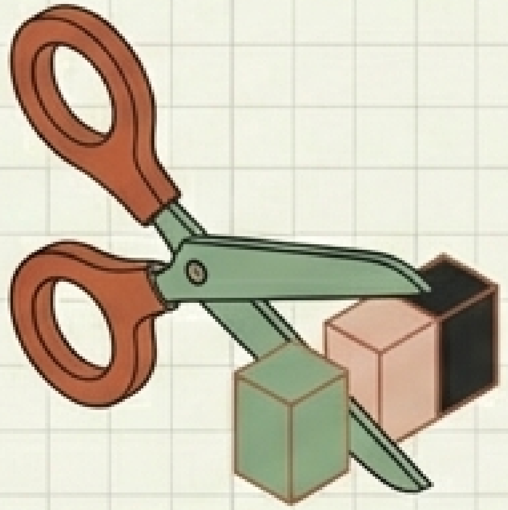
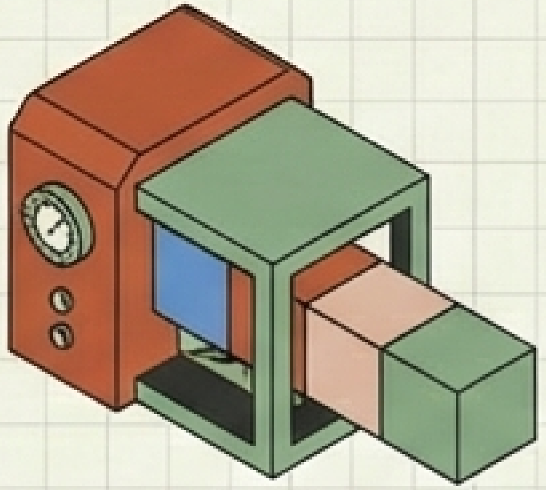
Techniques:

1. PCA (Principal Component Analysis)
2. LDA (Linear Discriminant Analysis)
3. t-SNE

Characteristic:

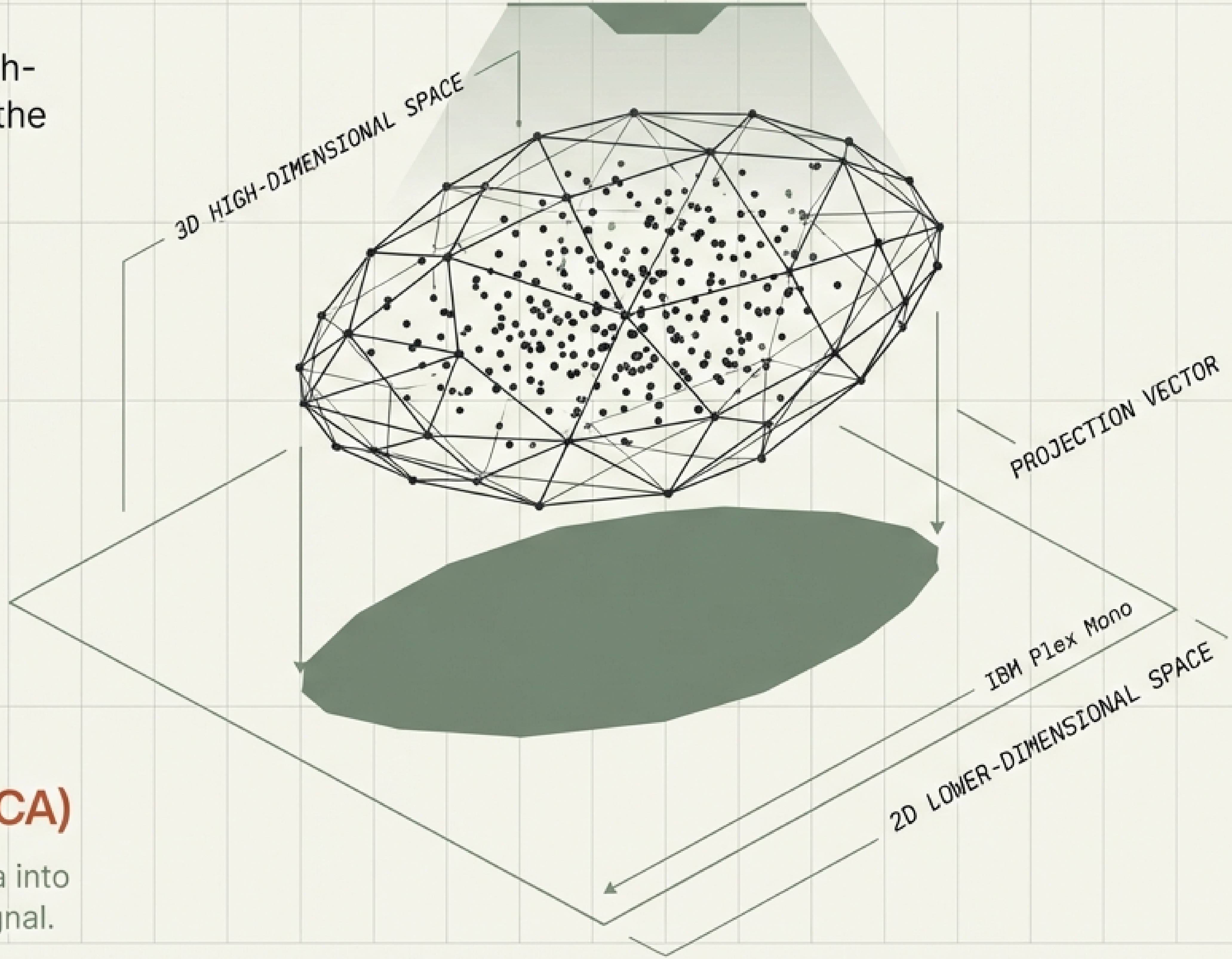
New features are linear combinations of old ones. Mathematically powerful, but less interpretable.

Selection vs. Extraction: Which Path to Take?

Feature Selection 		Feature Extraction 	
ACTION:	Subset of original columns.	ACTION:	Projection into new space.
PRO:	Keeps original meaning; easy to explain to stakeholders.	PRO:	Compresses data efficiently; handles complex correlations well.
CON:	Might miss complex relationships or combinations between features.	CON:	'Black box' features. Loss of physical interpretability.

Next Steps: Mastering Principal Component Analysis

Now that we understand the danger of high-dimensional space, we are ready to apply the most popular algorithm for fixing it.



COMING NEXT:
Principal Component Analysis (PCA)

Mathematically projecting high-dimensional data into a lower-dimensional space without losing the signal.